# ESKIŞEHIR TECHNICAL UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

BIM437 THESIS PROPOSAL FORM

# BIM437 THESIS PROPOSAL FORM

## A. GENERAL INFORMATION

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|---|
| **Applicant's Full Name:Enis Çelik, Furkan Çimili, Ahmet Arif Sarı** |
| **Title of the Thesis Proposal: Bank Customer Churn Prediction Using Machine Learning Techniques** |
| **Advisor's Full Name: Dr.Öğr.Üy.Gökhan Göksel** |
| **Institution/Organization Where the Thesis Will Be Conducted: Eskişehir Technical University, Department of Computer Engineering** |

**ABSTRACT**

The thesis proposal abstract is expected to include information on: (1) Scientific merit; (2) Methodology, (3) Project management; (4) Broader impact. It is recommended that this section be written last.


**Abstract**

This thesis proposal delineates a sophisticated machine learning framework designed to forecast customer churn in the banking industry, tackling a critical economic issue characterized by attrition rates of 10-30% that impose substantial revenue deficits. From a scientific standpoint, the project holds merit by rectifying deficiencies in model interpretability and equity observed in prior research, employing ensemble algorithms such as XGBoost and explainable AI methodologies like SHAP to deliver enhanced precision while alleviating biases in inherently imbalanced datasets. The pivotal research inquiry is: "In what ways can machine learning methodologies be refined to anticipate bank customer churn, concurrently upholding principles of algorithmic equity, environmental sustainability, and operational deployability?" Hypotheses include: (1) Ensemble approaches like XGBoost will surpass foundational classifiers in managing data imbalances, yielding a minimum 10% uplift in F1-score owing to their adeptness at discerning non-linear patterns; (2) The incorporation of SHAP will diminish bias by as much as 15%, quantifiable through fairness indicators like disparate impact, thereby fortifying ethical congruence sans performance degradation. This endeavor harmonizes with Turkey's 12th Development Plan (2024-2028) and 2030 Industry and Technology Strategy, which prioritize AI and data analytics to propel fintech innovations, cultivate a robust digital economy, and diminish reliance on imports through homegrown AI solutions and blockchain synergies, ultimately fostering competitive edges and potentially curtailing industry-wide churn expenditures that annually erode billions in earnings. Methodologically, the investigation adopts a rigorous pipeline encompassing: acquisition of datasets from Kaggle repositories (e.g., encompassing 10,000 records with attributes including age, account balance, and churn status); exploratory data analysis utilizing Pandas and Seaborn for correlation insights; preprocessing via OneHotEncoder for categorical encoding, StandardScaler for normalization, and SMOTE for imbalance rectification; model development and refinement through Scikit-learn with GridSearchCV for hyperparameter optimization, drawing on empirical evidence affirming XGBoost's efficacy in churn scenarios (Sahin et al., 2024); assessment employing metrics such as F1-score and ROC-AUC; and deployment facilitated by Flask APIs, Streamlit interfaces, and Docker containerization to ensure scalability. Architectural blueprints are articulated via the C4 model (Context, Container, Component, Code levels) and rendered using Structurizr DSL, substantiating a multifaceted system architecture. Initial feasibility assessments on data subsets have demonstrated 84% accuracy, seamlessly aligned with the project's six-month chronology.
In terms of project management, a segmented timeline spans November 2025 to April 2026, featuring milestones like EDA in the inaugural month (contributing 20% to overall success) and deployment in Month 4 (20%), culminating in a 100% aggregate of quantifiable success benchmarks. Potential hazards, including data skew, are countered through contingencies such as ADASYN resampling, by using our computers and corporate assets like our software packages like Jupyter and GitHub.
The broader ramifications encompass scholarly contributions (e.g., a prospective IEEE conference publication), socioeconomic advantages (a functional web prototype ameliorating financial disparities via equitable AI), and catalysts for subsequent initiatives (e.g., TUBITAK 2209-B expansions targeting real-time analytics). This initiative resonates with UN Sustainable Development Goals 8 (Decent Work and Economic Growth), 9 (Industry, Innovation, and Infrastructure), and 10 (Reduced Inequalities), advocating for inclusive financial ecosystems and innovative paradigms.



**Keywords:** Customer Churn Prediction, Machine Learning, XGBoost, Explainable AI, Banking Analytics


## 1. SCIENTIFIC MERIT OF THE THESIS PROPOSAL

### 1.1. Significance of the Topic and Scientific Merit of the Thesis Proposal

## BIM437 THESIS PROPOSAL FORM

In the thesis proposal, the scope, boundaries, and significance of the selected topic are presented. The proposal should clearly explain, with reference to the relevant literature, which gap in the existing research will be addressed or which problem will be solved through the proposed study, thereby demonstrating the scientific merit of the research. The research question and, if applicable, the hypothesis (or hypotheses) are defined.

The thesis topic should be related to the critical technology areas or the priority R&D and innovation fields identified in the *12th Development Plan* and the *2030 Industry and Technology Strategy*, the reason for this connection and the potential benefits the study will provide to the relevant area should be explained.

In addition, the thesis topic must include the following criteria:

*(i)* A complex problem is "a comprehensive problem that requires some or all of the following elements for its solution: in-depth engineering knowledge, abstract thinking, creative use of fundamental engineering principles and research-based knowledge in leading areas of the computer engineering discipline, and development of a new model or method; concerns a variety of stakeholders with different needs; and may have significant consequences in a variety of contexts."
*The thesis must address a complex problem:* the definition of the complex engineering problem of the thesis using fundamental science, mathematics, and engineering knowledge, while considering the relevant United Nations Sustainable Development Goals (https://sdgs.un.org/goals ) must be given.

*(ii)* The realistic constraints and conditions in engineering design are depending on the nature of the design, these are elements such as economy, environmental problems, sustainability, manufacturability, ethics, health, safety, social, time, performance, functionality, legal and political dimensions.
*The thesis must address realistic constraints and conditions in engineering design:* the realistic constraints and conditions that need to be taken into account during the design.

In the highly competitive banking sector, customer churn represents a pervasive challenge, with annual attrition rates typically ranging between 10% and 30%, resulting in substantial economic repercussions. Research indicates that the cost of acquiring new customers can be 5 to 25 times higher than retaining existing ones, underscoring the imperative for proactive retention strategies. This thesis proposal seeks to bridge a critical gap in the extant literature by devising an advanced predictive model that not only attains superior accuracy through ensemble machine learning techniques but also embeds explainable artificial intelligence (XAI) mechanisms, such as SHAP (SHapley Additive exPlanations), to address ethical concerns like algorithmic bias and transparency—issues often sidelined in conventional models. For instance, while prior studies have achieved promising results with models like Random Forest and XGBoost in churn prediction, they frequently overlook the interpretability required for regulatory compliance and stakeholder trust, particularly in imbalanced datasets prevalent in banking contexts.
The central research question guiding this study is: "How can machine learning techniques be optimized to predict bank customer churn while ensuring algorithmic fairness, sustainability, and real-world deployability?" To explore this, the following hypotheses are posited: (1) Ensemble models, such as XGBoost, will outperform baseline classifiers (e.g., Logistic Regression) in handling imbalanced datasets, achieving at least a 10% improvement in F1-score due to their robustness in capturing non-linear relationships; (2) Integrating XAI tools like SHAP will reduce model bias by up to 15%, as measured by fairness metrics such as disparate impact, thereby enhancing ethical alignment without compromising predictive performance.
This topic aligns seamlessly with the critical technology areas and priority R&D and innovation fields outlined in Turkey's 12th Development Plan (2024-2028) and the 2030 Industry and Technology Strategy. Specifically, the plan emphasizes artificial intelligence (AI) and data analytics as pivotal for financial innovation, aiming to foster a digital economy through indigenous AI development, fintech advancements, and blockchain integration to enhance competitiveness and reduce import dependency. By leveraging AI-driven predictive analytics, this thesis contributes to these national priorities by potentially mitigating sector-wide churn costs—estimated to impact billions in revenue annually—and promoting a resilient digital financial ecosystem. This could accelerate Turkey's goals of achieving 5.0% annual real GDP growth, elevating high-tech exports from 3% to 17% of total exports, and positioning the country as a regional hub for fintech innovation, including secure digital currencies and efficient capital markets via the Istanbul Financial Center.
Furthermore, the thesis tackles a complex engineering problem as defined in the program's guidelines: It necessitates profound engineering knowledge in algorithms and data structures for model optimization, abstract reasoning to manage imbalanced data (e.g., employing SMOTE oversampling grounded in statistical probability distributions), innovative application of core engineering principles such as Bayesian probability in classification tasks, and cutting-edge research in machine learning domains like gradient boosting and neural networks. The problem engages diverse stakeholders, including customers (demanding data privacy), banking institutions (seeking economic advantages), regulatory bodies (enforcing compliance), and broader society (benefiting from economic stability), with potential ramifications such as exacerbated financial inequalities if biases remain

unchecked or enhanced inclusivity through fair AI. Grounded in fundamental sciences—statistics for model evaluation, mathematics for optimization via gradient descent, and engineering knowledge in scalable software design patterns—this work also integrates relevant United Nations Sustainable Development Goals (SDGs): SDG 9 (Industry, Innovation, and Infrastructure) by advancing AI for streamlined banking operations and digital infrastructure; SDG 8 (Decent Work and Economic Growth) via improved customer retention strategies that bolster financial sector employment and productivity; and SDG 10 (Reduced Inequalities) through bias-mitigated models that promote equitable access to financial services, aligning with global efforts to minimize algorithmic discrimination.
The thesis also rigorously incorporates realistic constraints and conditions inherent to engineering design: Economic factors, such as low-cost cloud deployment using platforms like Google Colab to minimize infrastructure expenses; environmental considerations, including energy-efficient model architectures (e.g., lightweight neural networks) to reduce carbon footprints associated with computational training; sustainability through reusable, modular code frameworks that facilitate long-term maintenance and scalability; manufacturability via Python-based implementations that are easily deployable across standard hardware; ethical imperatives, encompassing data anonymization techniques and fairness audits to prevent discriminatory outcomes; health and safety aspects, such as averting customer stress from erroneous predictions by prioritizing recall in high-risk churn scenarios; social dimensions, promoting inclusive finance for underserved populations; time constraints, ensuring real-time API responses under 1 second for practical usability; performance requirements, targeting accuracy exceeding 85% as benchmarked against literature; functionality in the form of a user-friendly web application interface; legal compliance with regulations like GDPR and Turkey's KVKK for data protection; and political alignment with national digital policies to support Turkey's vision of a knowledge-based economy.

### 1.2. Objectives and Goals

The purpose and objectives of the thesis proposal should be written in a clear, measurable, realistic, and achievable manner.

The primary purpose is to develop a robust ML system for bank customer churn prediction that is accurate, ethical, and deployable. Specific objectives include: (1) Analyze and preprocess imbalanced datasets from Kaggle (e.g., 10,000 rows with 14 features) to achieve balanced classes measurable by F1-score >0.85; (2) Train and optimize ML models (Logistic Regression, Random Forest, XGBoost) using cross-validation, targeting ROC-AUC >0.90; (3) Implement explainable AI (SHAP) to interpret predictions, ensuring bias reduction quantifiable by fairness metrics (e.g., disparate impact <1.2); (4) Deploy a web application with Flask or Streamlit for real-time predictions, testable with user simulations; (5) Evaluate against benchmarks from literature (e.g., 86% accuracy in similar studies), realistic within 6 months using university resources. These goals are achievable with team collaboration and weekly advisor meetings, aligning with ABET criteria for complex engineering solutions.

## 2. METHODOLOGY

The suitability of the methods and research techniques to be applied in the thesis study for achieving the stated aims and objectives should be demonstrated with references to the relevant literature.
The methodology section should include elements such as the design of the thesis study, dependent and independent variables, statistical methods, and other relevant components. If any preliminary study or feasibility analysis has been conducted, it is expected to be presented within the proposal. Furthermore, the methods proposed in the thesis should be linked to the research timeline to ensure coherence between the methodology and the project schedule.

*The thesis must include architecture diagrams:* The architecture diagrams prepared using the C4 model (Context, Container, Component, Code), as detailed at https://c4model.com/. These diagrams must be rendered using the Structurizr DSL (https://structurizr.com/dsl/).

The methodology employs a structured ML pipeline: (1) Data collection from Kaggle datasets (e.g., Bank Customer Churn Dataset with features like Age, Balance, Exited); (2) Exploratory Data Analysis (EDA) using Pandas and Seaborn for correlations; (3) Preprocessing with OneHotEncoder for categorical variables, StandardScaler for scaling, and SMOTE for imbalance; (4) Model selection and training via Scikit-learn (Random Forest, XGBoost) with GridSearchCV for hyperparameter tuning, supported by literature showing XGBoost's superiority in churn tasks (Sahin et al., 2024); (5) Evaluation using metrics like F1-score, ROC-AUC; (6) Deployment with Flask API,Docker and Streamlit for scalability. Preliminary feasibility: Tested on sample data yielding 84% accuracy. Timeline integration: EDA in Month 1, training in Month 2-3.
The thesis includes architecture diagrams using the C4 model:

Context: System overview – Bank Churn Prediction Platform interacts with users (bank staff) and external data

sources (datasets/APIs).
Container: Data Storage (SQL), ML Server (Python Flask), User Interface (Web Frontend).
Component: Data Pipeline Component (preprocessing), Prediction Component (model inference), Visualization Component (SHAP explanations).
Code: Class details, e.g., ChurnPredictor class with methods for train() and predict().

**Structurizr DSL Code :**

```
workspace "Bank Churn Prediction" {
  model {
    user = person "Bank Staff" "Uses the system to predict churn."
    system = softwareSystem "Churn Prediction Platform" "Predicts customer churn using ML." {
      db = container "Database" "Stores customer data" "SQLite"
      mlServer = container "ML Server" "Runs models" "Python Flask" {
        dataPipeline = component "Data Pipeline" "Preprocesses data"
        predictor = component "Predictor" "Runs ML model"
        visualizer = component "Visualizer" "Generates explanations"
        dataPipeline -> predictor "Feeds processed data"
        predictor -> visualizer "Sends results"
      }
      ui = container "Web UI" "User interface" "HTML/JS"
      user -> ui "Interacts with"
      ui -> mlServer "Calls API"
      mlServer -> db "Queries data"
    }
    user -> system "Accesses predictions"
  }
  views {
    systemContext system "ContextView" "Overall system context." {
      include *
      autoLayout
    }
    container system "ContainerView" "High-level containers." {
      include *
      autoLayout
    }
    component mlServer "ComponentView" "ML Server components." {
      include *
      autoLayout
    }
     theme Default
  }
}
```

## 3. PROJECT MANAGEMENT

### 3.1 Work Schedule

The main activities to be carried out within the scope of the thesis study, the duration of each activity, the criteria for success, and their contribution to the overall success of the research should be presented by completing the “Work Schedule” section.

**WORK SCHEDULE (*)**

| Date Range | Activities ** | Responsible Person(s) | Success Criteria and Contribution to the Success of the Research*** |
|---|---|---|---|
| 01/11/2025-30/11/2025 | Dataset collection and EDA | All team members | Complete preprocessing with imbalance handled |

| | | | |
|---|---|---|---|
| | | | (F1-score baseline >0.70); Contributes 20% |
| 01/12/2025-31/01/2026 | Model training and optimization | All team members | Achieve ROC-AUC >0.85 on validation set; Contributes 30% (core prediction accuracy). |
| 01/02/2026-28/02/2026 | Explainability integration and testing | All team members | Bias metrics <1.2, SHAP visualitzations generated; Contributes 20% (ethical compliance). |
| 01/03/2026-31/03/2026 | Web app development and deployment | All team members | Functional API with <1s response time, tested on 100 samples; Contributes 20% (usability). |
| 01/04/2026-30/04/2026 | Evaluation and report finalization | Advisor + team | Comparative analysis with literature, final accuracy > 85%; Contributes 10% (validation). |

(*) *The rows and columns in the table may be expanded or duplicated as needed.*
(****) The stages of literature review, preparation of the final report, dissemination of research results, and procurement of materials should not be presented as separate work steps.*
**(*****) The success criterion should be expressed using quantitative or qualitative indicators (such as statements, numbers, or percentages) that are measurable and trackable. The total value in this column must equal 100.

### 3.2 Risk Management

The risks that may negatively affect the success of the thesis and the precautionary measures (Plan B) to ensure the successful continuation of the study in the event of such risks should be presented in the *Risk Management Table* below. The implementation of Plan B should not lead to any deviation from the main objectives of the thesis. If the adoption of Plan B requires a change in the research methodology, this modification should be explained in detail.

**RISK MANAGEMENT TABLE ***

| Major Risks | Precautionary Reasons (Plan B) |
|---|---|
| Data imbalance leading to poor model performance | -Use ADASYN instead of SMOTE for adaptive synthetic sampling<br>-Implement cost-sensitive learning with class weights (weight = n_samples / (n_classes × n_samples_class))<br>- Use ensemble methods (Balanced Random Forest) |
| Hardware limitations for training (insufficient RAM/CPU) | -Shift to cloud (Google Colab) – maintains objectives.<br>-Use university computer lab workstations<br>- Reduce hyperparameter search space (from 200 to 50 iterations) |
| Bias in model predictions (fairness violations) | -Integrate Fairlearn library for mitigation<br>-Implement threshold optimization per demographic group<br>-Use adversarial debiasing techniques |
| Dataset quality issues (missing values >20%, data leakage) | -Switch to alternative Kaggle dataset (Telco Churn - 7K records)<br>-Use multiple imputation methods (KNN, Iterative)<br>-Synthetic data generation (CTGAN) if necessary |

| | |
|---|---|
| Model performance below target (F1 <0.85) | -Deep feature engineering (30+ interaction features)<br>-Advanced ensemble (Stacking with 5 base models)<br>-Redefine success criteria with business justification |

(*) The rows in the table may be expanded or duplicated as needed.

### 3.3.Research Facilities

This section should specify the available infrastructure and equipment (such as laboratories, tools, machinery, and technical devices) at the institutions or organizations where the thesis will be conducted, which will be utilized during the course of the project.

**RESEARCH FACILITIES TABLE (*)**

| **Type/Model of Infrastructure/Equipment at the Institution**<br>(Laboratory, Vehicle, Machinery-Equipment, etc.) | **Purpose of Use in the Project** |
|---|---|
| Personal Computers | -Primary development environment for Jupyter Notebooks, VS Code, model training (small-scale), local API testing |
| Cloud Platform | -Training computationally expensive models (XGBoost, Neural Networks), hyperparameter tuning with GridSearchCV, team collaboration |
| Software - Python 3.9+ | -Libraries: pandas, numpy, scikit-learn, xgboost, shap, flask, streamlit, docker<br><br>-Data preprocessing, ML modeling, explainability (SHAP), API development, containerization |
| Eskisehir Technical University Computer Engineering Labs | -High-performance workstations<br><br>-Backup computing resources, presentation preparation, access to university network resources |
| Version Control - GitHub | -Repository: github.com/[team-name]/bank-churn-prediction<br><br>- Source code management, collaboration, version tracking, CI/CD pipeline, documentation hosting |
| Development Tools | -Jupyter Notebook, VS Code, Git, Anaconda<br><br>-Interactive data analysis, production code development, environment management |

(*) The rows in the table can be expanded or duplicated as needed.

**4. BROADER IMPACT OF THE THESIS PROPOSAL**

The expected outputs from the work proposed in the thesis should be categorized according to their objectives and clearly specified. These outputs must be based on measurable and realistic targets.

| Output, Impact, and Benefits | Expected Outputs, Impacts, and Benefits |
|---|---|
| **Scientific/Academic Outputs**<br>(National/International Articles, Book Chapters, Books, Conference Papers, etc.) | -1 conference paper submission to IEEE Signal Processing and Communications Applications Conference (SIU 2026) by April 2026<br>-Thesis Report: 35-40 pages comprehensive document serving as reference for future ML churn prediction studies<br>-GitHub Repository: Open-source codebase with 100+ stars target within 6 months |
| **Economic/Commercial/Social Outputs**<br>(Product, Prototype, Patent, Utility Model, Registration, Audio/Visual Archive, Inventory/Database, Workshop, Training, Scientific Event, etc.) | -Deployable Web Prototype: Functional Flask API + Streamlit dashboard accessible via public URL<br>-Economic Impact: Framework demonstrates potential savings of $12M+ annually for mid-sized banks (500K customers, 20% churn rate, $1,200 CLV)<br>-Social Impact: Bias mitigation promotes equitable financial services, reducing algorithmic discrimination by 15% (measured via disparate impact ratio)<br>-Industry Collaboration: Potential partnership with 1 Turkish bank for pilot implementation (target: İş Bankası, Garanti BBVA, or Yapı Kredi) \| |
| **Outputs for Generating New Project(s)**<br>(National/International New Projects, etc.) | -TÜBİTAK 2209-B Extension: Industry-oriented version with real bank partner (Spring 2027 application)<br>-Future Research: Real-time churn prediction with streaming data (Kafka integration), temporal analysis with LSTM networks<br>-Career Impact: Portfolio project demonstrating full-stack ML skills for job applications |

**5. OTHER TOPICS YOU WISH TO MENTION**

Only information or data (such as graphs, tables, etc.) that can contribute to the evaluation of the thesis proposal may be included.

5.1. Innovative Aspects of the Thesis

This thesis distinguishes itself through:

**1. Dual Focus on Performance and Interpretability**: Unlike most studies that prioritize accuracy alone, we integrate SHAP-based explainability from the outset, addressing the critical gap identified in recent literature where high-accuracy models fail adoption due to lack of transparency.

**2. Comprehensive Bias Testing:** Systematic fairness auditing across demographic groups (Age, Gender, Geography) using multiple metrics (disparate impact, equal opportunity), going beyond standard ML practices.

**3. Production-Ready Architecture:** Full-stack implementation with REST API, database integration, and containerization (Docker), bridging the research-to-practice gap.

5.2. Preliminary Results

Initial experiments on 1,000 sample records demonstrate:

- Baseline Logistic Regression: 84% accuracy, 0.76 AUC-ROC
- Confirmed class imbalance: 80.2% retained, 19.8% churned (manageable)
- Key insight: Age and IsActiveMember show strong correlation with churn ($p < 0.001$)
- Training time: <3 seconds (confirms computational feasibility)

[ATTACHED GRAPH: Pie chart showing churn distribution - 20% churned, 80% retained]

[ATTACHED TABLE: Feature correlation matrix showing Age (r=0.28), NumOfProducts (r=-0.35)]

5.3. Alignment with UN Sustainable Development Goals
Beyond SDG 8, 9, and 10 mentioned in Section 1:
- SDG 4 (Quality Education): Open-source GitHub repository serves as educational resource for future students learning ML applications

- SDG 12 (Responsible Production): Reduces wasteful customer acquisition spending, promoting sustainable business practices

5.4. Team Expertise and Collaboration Our team brings complementary skills:
- Enis Çelik: Strong in ML theory and model development
- Furkan Çimili: Experienced in software engineering and API development
-*Ahmet Arif Sarı: Skilled in system architecture and technical documentation

Weekly meetings with advisor Dr. Gökhan Göksel ensure progress and quality control.

5.5. Broader Applicability
The developed framework is adaptable to:
- Telecommunications (mobile/internet service churn)
- E-commerce (customer retention)
- SaaS platforms (subscription cancellation)
- Healthcare (patient attrition from treatment programs)

This demonstrates the generalizability of our approach beyond banking.

**6. REFERENCES**

All references must be written in APA-style.

Almaliki, S., Alghamdi, A., & Alharbi, A. (2023). Investigating customer churn in banking: A machine learning approach and visualization app for data science and management. *Data Science and Management*, *6*(3), Article 100040. https://doi.org/10.1016/j.dsm.2023.09.002

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Yadav, A., et al. (2024). Improving churn detection in the banking sector: A machine learning approach with probability calibration techniques. *Electronics*, *13*(22), 4527.